How-To Library

Informatica

Transferring data between Kafka topics and MongoDB collections in Cloud Data Integration

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Abstract

This article provides detailed information on how you can configure a mapping to transfer data between Kafka topics and MongoDB collections in Data Integration.

Supported Versions

- Cloud Data Integration Kafka Connector Spring 2020 April

Table of Contents

Overview. 2
Prerequisites. 2
Reading data from a Kafka topic and writing to a MongoDB collection. 2
Reading data from a MongoDB collection and writing to a Kafka topic. 4

Overview

You can configure a mapping to transfer data between Kafka topics and MongoDb collections in JSON format. You can read data from Kafka in JSON format and write to a MongoDB collection. You can also read data from a MongoDB collection and write data to a Kafka topic in JSON format.

Prerequisites

Before you configure a mapping to transfer data between Kafka topics and MongoDB collections, you must complete the following prerequisites:

- Ensure that you have the Kafka Connector and MongoDB Connector licenses enabled for your organization. For more information, contact Informatica Global Customer Support.
- Ensure that you have created the Kafka and MongoDB connections in the Administrator service.

Reading data from a Kafka topic and writing to a MongoDB collection

Configure a mapping in Data Integration to read data from a Kafka topic and write to the DocumentAsJson field in the MongoDB target.

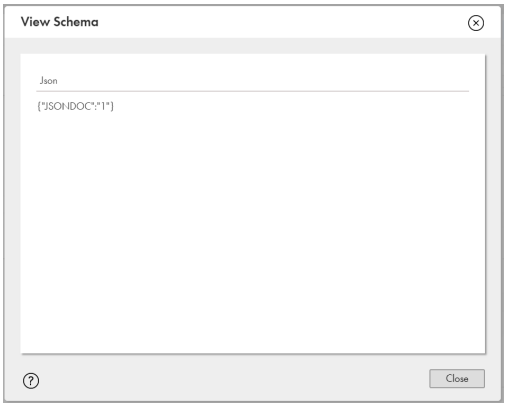
Perform the following tasks to configure a mapping:

1. Click **New** > **Mappings**, and then click **Create**.

   The Mapping Designer appears with a Source transformation and a Target transformation in the mapping canvas.
2. In the Source transformation, click the **Source tab** and select a Kafka connection as the source connection.
3. Select a Kafka topic as the source object, and click **Formatting Options**.

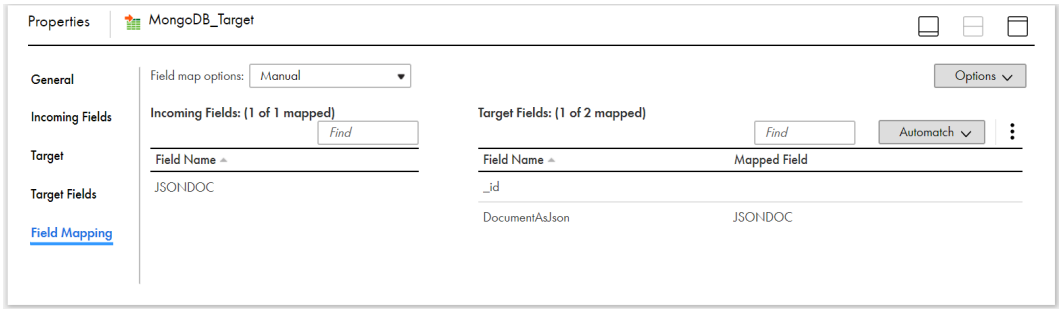
   The Formatting Options dialog box opens.
4. In the **Format Type** option, select JSON.
5. In the **Schema Source** option, select **Import from Schema File**.

You must specify a JSON schema with a single column in the schema file.
For example, the following image shows a JSON schema configured for the Kafka source object:



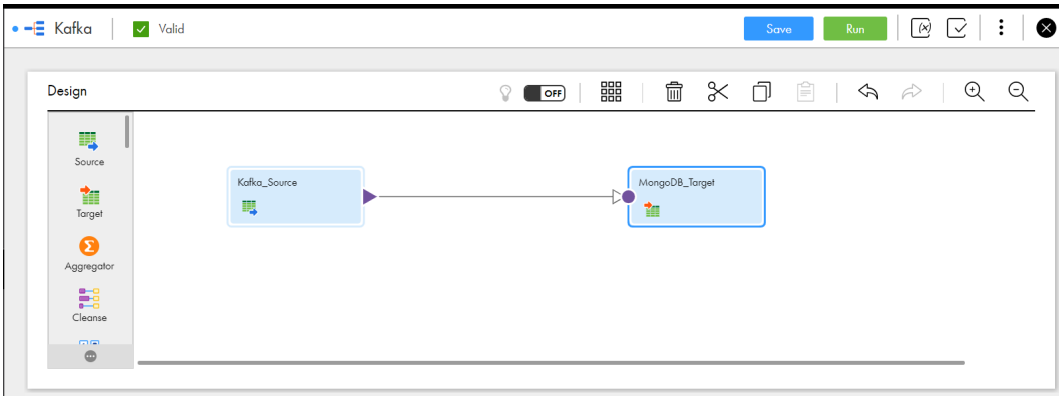
6. In the Target transformation, click the **Target** tab and select a MongoDB connection as the source connection.
7. Select a MongoDB collection as the target object, and select Insert as the **Operation**.
8. Click **Field Mapping** and map the JSON data field in the Kafka source to the DocumentAsJson field in the MongoDB target.

   The following image shows the mapped JSONDOC field in the Kafka source to the DocumentAsJson field in the MongoDB target:



9. To save your changes and continue, click **Save**.
10. Click **Run** to run the mapping.

    The following image shows the configured mapping:



When you run the mapping, the Secure Agent writes the data from the JSONDOC field in the Kafka source to the DocumentAsJSON field in the MongoDB target.

Reading data from a MongoDB collection and writing to a Kafka topic

Configure a mapping in Data Integration to read data from a MongoDB collection and write to a Kafka topic. You can read data from all the columns in the MongoDB collection through the DocumentAsJson column and write to the Kafka target. If you want to read data from individual columns in the MongoDB collection and write to the Kafka target, you must specify the JSON schema based on the columns that you want to read from the MongoDB source object.

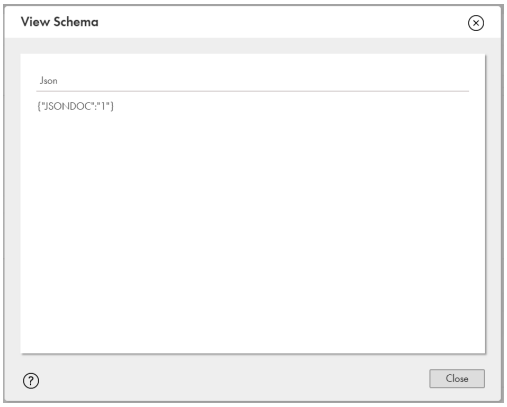
Perform the following tasks to configure a mapping:

1. Click **New** > **Mappings**, and then click **Create**.

   The Mapping Designer appears with a Source transformation and a Target transformation in the mapping canvas.
2. In the Source transformation, click the **Source** tab and select a MongoDB connection as the source connection.
3. Select a MongoDB collection as the source object.
4. In the Target transformation, click the **Target** tab and select a Kafka connection as the source connection.
5. Select a Kafka topic as the target object, and click **Formatting Options**.

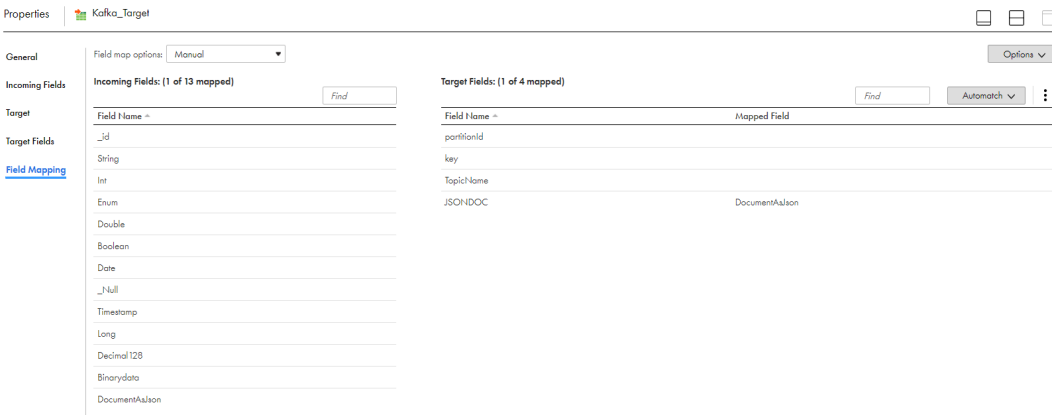
   The Formatting Options dialog box opens.
6. In the **Format Type** option, select JSON.
7. In the **Schema Source** option, select **Import from Schema File**.

   If you want to read data from all the columns in the MongoDB collection through the DocumentAsJson column and write to the Kafka target, you must specify a JSON schema with a single column in the schema file for Kafka target object. If you want to read data from individual columns in the MongoDB collection and write to the Kafka target, you must specify the JSON schema based on the columns that you want to read from the MongoDB source object.
   The following image shows a JSON schema configured for the Kafka target object:

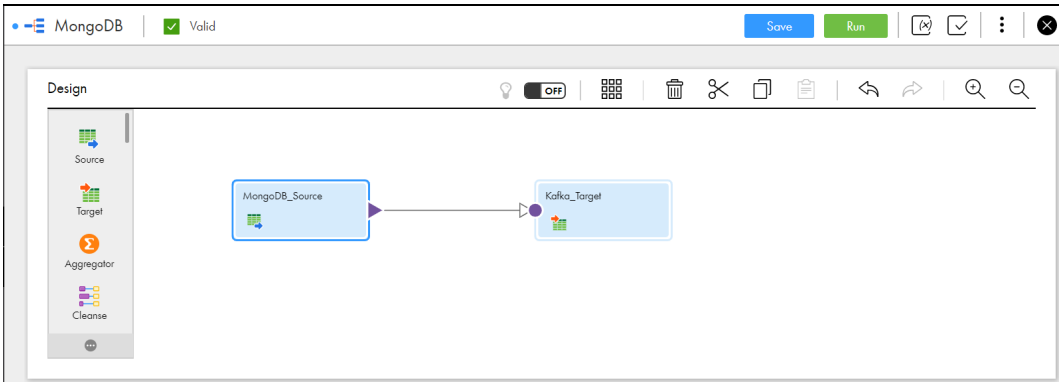
8. Select Insert as the **Operation**.
9. Click **Field Mapping** and map the DocumentAsJson field in the MongoDB source to the JSON data field in the Kafka target.

   The following image shows the mapped DocumentAsJson field in the MongoDB source to the JSONDOC field in the Kafka target:



10. To save your changes and continue, click **Save**.
11. Click **Run** to run the mapping.

    The following image shows the configured mapping:



    When you run the mapping, the Secure Agent writes the data from the DocumentAsJSON field in the MongoDB source to the JSONDOC field in the Kafka target.

Author

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Acknowledgements

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